

UESTCHN3007 Dynamics & Control  
Lecture 3

# Block Diagram Reduction

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# Outline

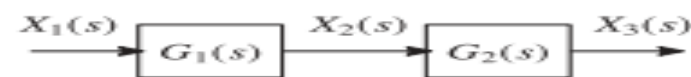
- Apply Block Diagram Reduction Rules
- Analyse Control System Subjected to a Disturbance

## Rule

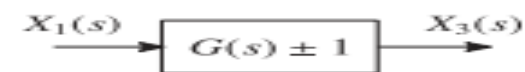
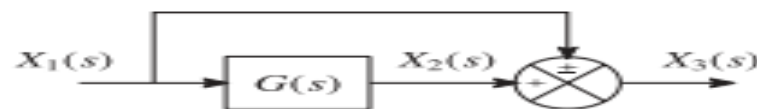
## Original

## Equivalent

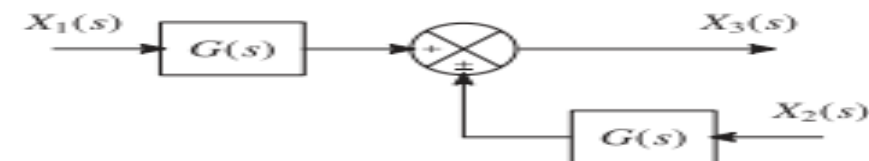
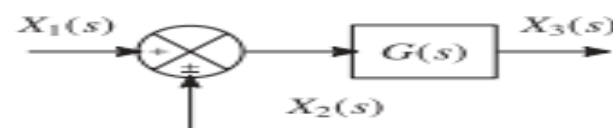
## 1. Cascaded blocks



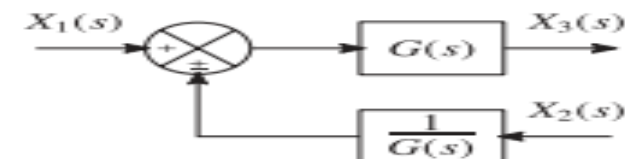
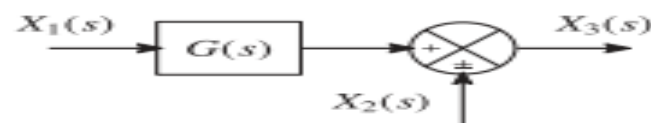
## 2. Summing two signals



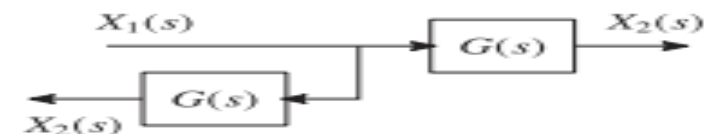
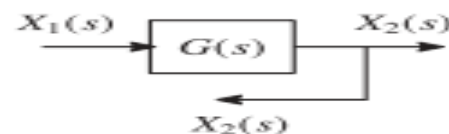
## 3. Moving a summing point behind a block



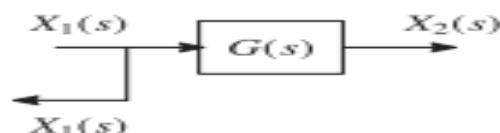
## 4. Moving a summing point ahead of a block



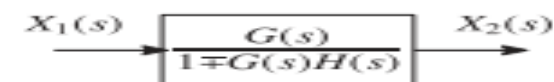
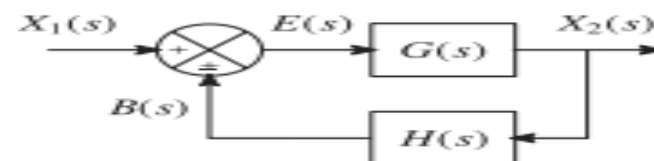
## 5. Moving a branch point ahead of a block



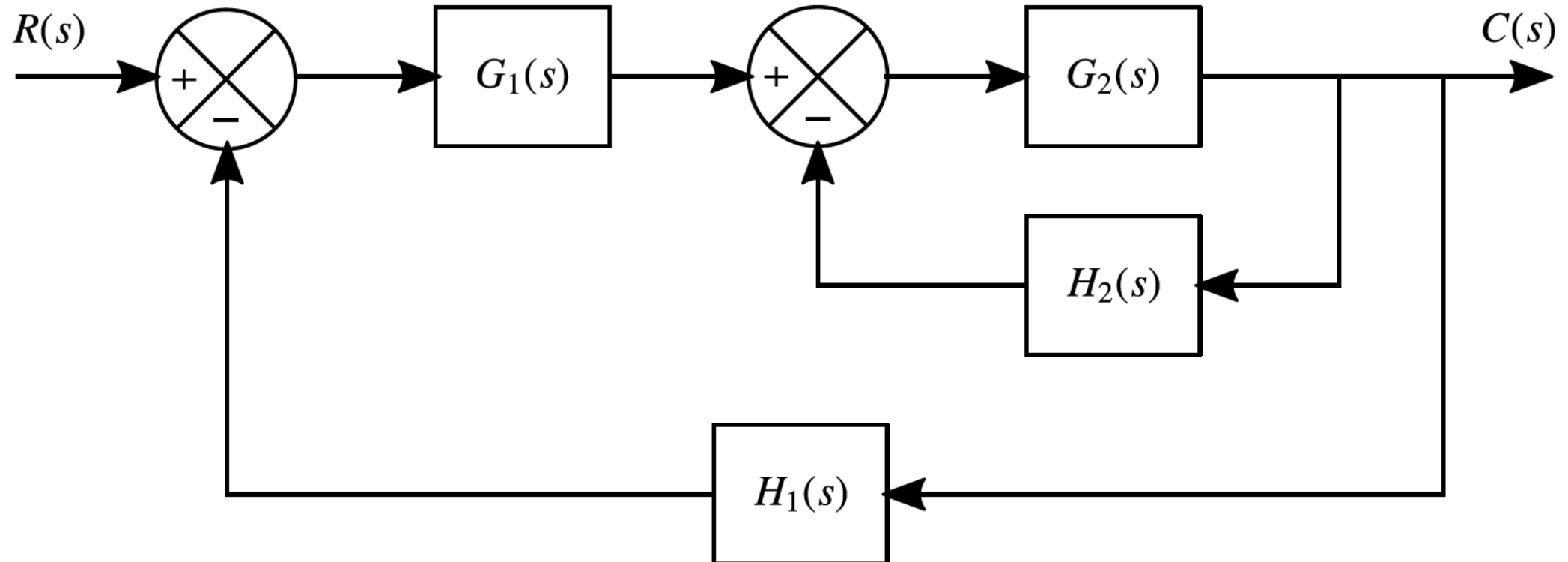
## 6. Moving a branch point behind a block



## 7. Eliminating a feedback loop



# Example: Derive the overall transfer function



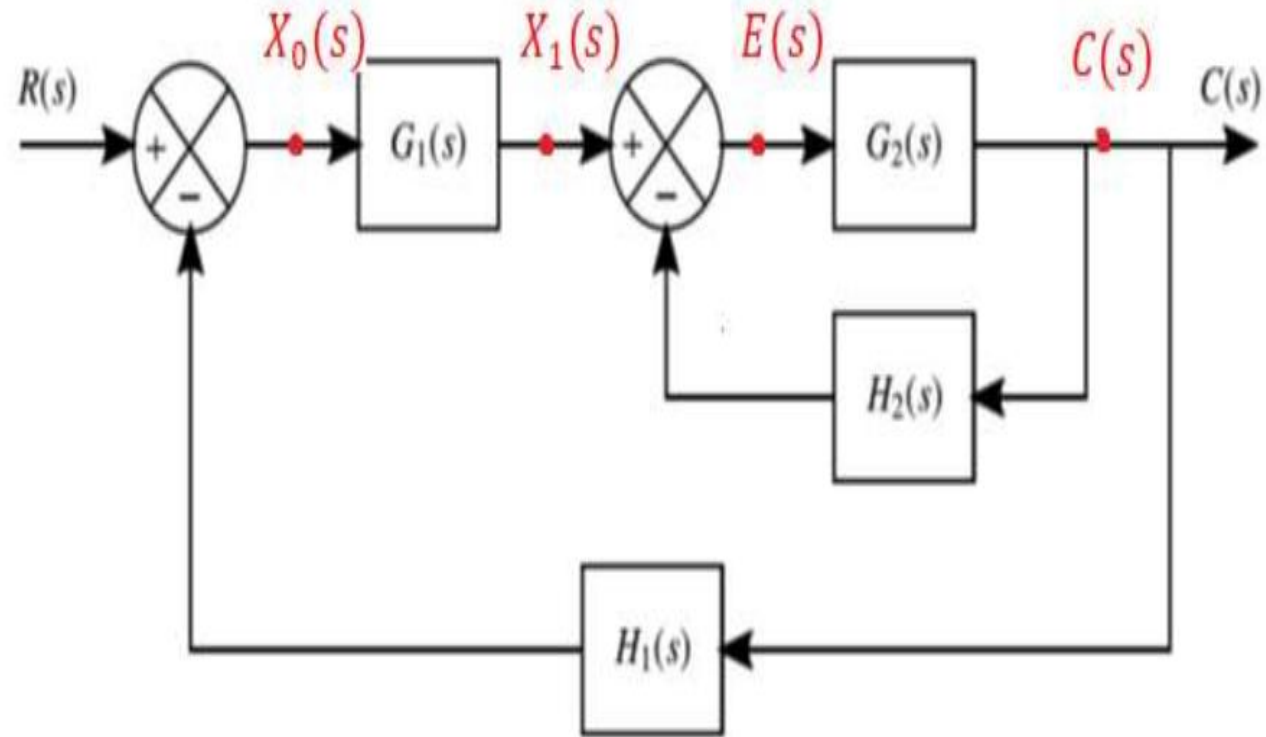
# Example: Derive the overall transfer function

$$C(s) = E(s)G_2(s) \quad (1)$$

$$E(s) = X_1(s) - C(s)H_2(s) \quad (2)$$

$$X_1(s) = X_0(s)G_1(s) \quad (3)$$

$$X_0(s) = R(s) - C(s)H_1(s) \quad (4)$$



Using (1)-(4):

$$C(s) = E(s)G_2(s)$$

$$= (X_1(s) - C(s)H_2(s))G_2(s)$$

$$= (X_0(s)G_1(s) - C(s)H_2(s))G_2(s)$$

$$= [(R(s) - C(s)H_1(s))G_1(s) - C(s)H_2(s)]G_2(s)$$

$$= R(s)G_1(s)G_2(s) - C(s)H_1(s)G_1(s)G_2(s) - C(s)H_2(s)G_2(s)$$

$$C(s) + C(s)H_1(s)G_1(s)G_2(s) + C(s)H_2(s)G_2(s) = R(s)G_1(s)G_2(s)$$

$$C(s) \cdot (1 + H_1(s)G_1(s)G_2(s) + H_2(s)G_2(s)) = R(s)G_1(s)G_2(s)$$

$$\frac{C(s)}{R(s)} = \frac{G_1(s)G_2(s)}{1 + H_1(s)G_1(s)G_2(s) + H_2(s)G_2(s)}$$

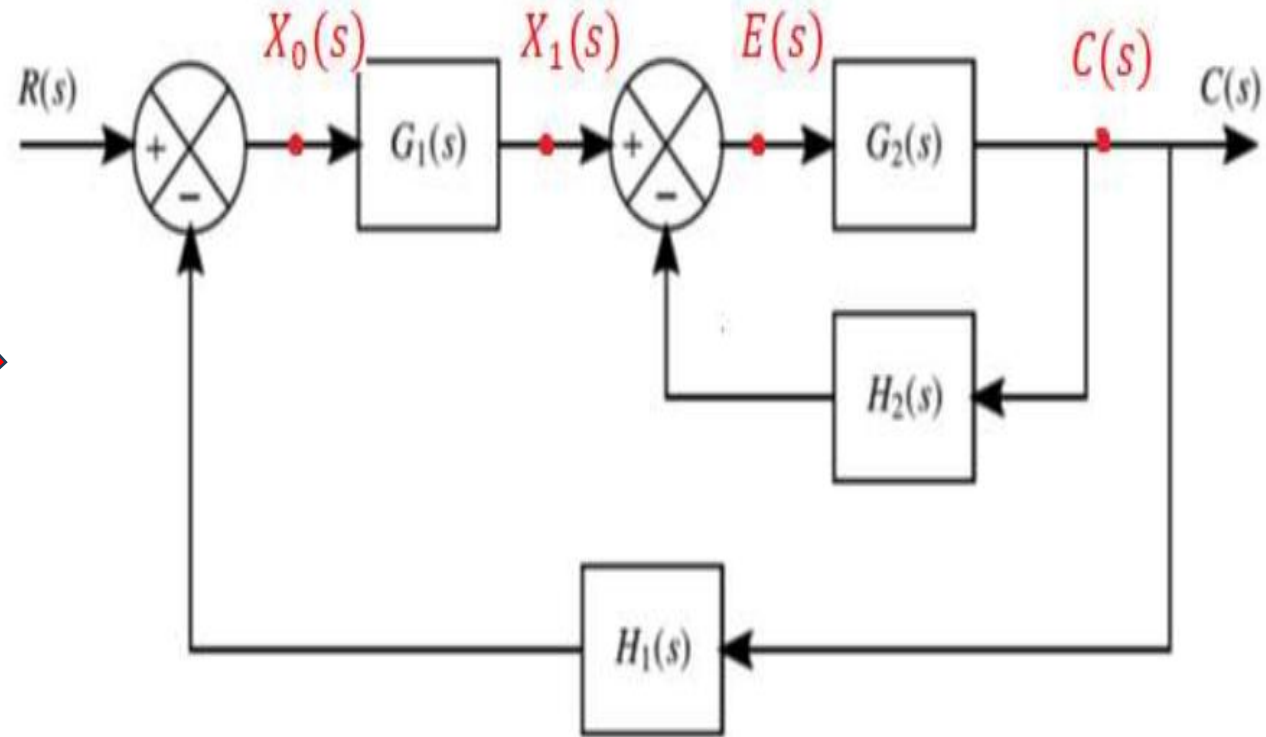
# Example: Derive the overall transfer function

$$C(s) = E(s)G_2(s) \quad (1)$$

$$E(s) = X_1(s) - C(s)H_2(s) \quad (2)$$

$$X_1(s) = X_0(s)G_1(s)$$

$$X_0(s) = R(s) - C(s)H_1(s)$$



Using (1)-(4):

$$C(s) = E(s)G_2(s)$$

$$= (X_1(s) - C(s)H_2(s))G_2(s)$$

$$= (X_0(s)G_1(s) - C(s)H_2(s)G_2(s))G_2(s)$$

$$= [(R(s) - C(s)H_1(s))G_1(s) - C(s)H_2(s)G_2(s)]G_2(s)$$

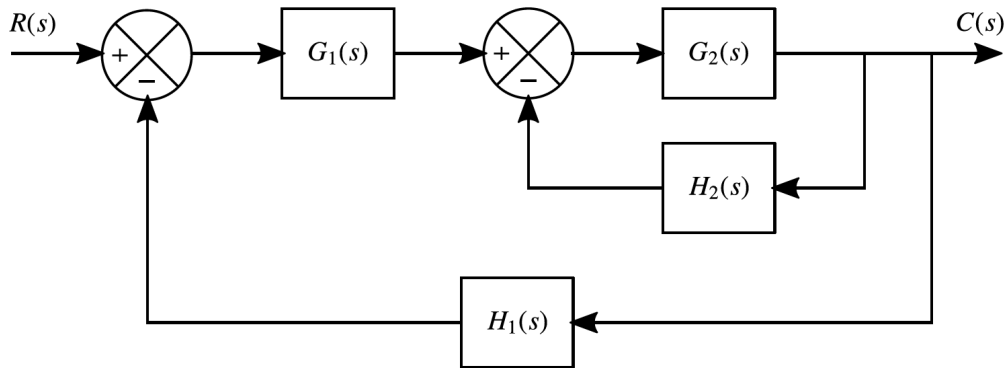
$$= R(s)G_1(s)G_2(s) - C(s)H_1(s)G_1(s)G_2(s) - C(s)H_2(s)G_2(s)$$

$$C(s) + C(s)H_1(s)G_1(s)G_2(s) + C(s)H_2(s)G_2(s) = R(s)G_1(s)G_2(s)$$

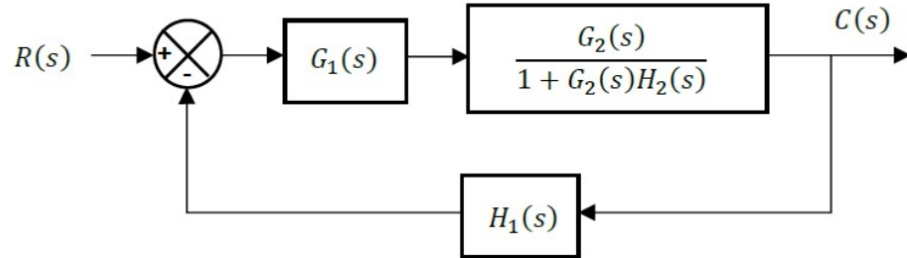
$$C(s) \cdot (1 + H_1(s)G_1(s)G_2(s) + H_2(s)G_2(s)) = R(s)G_1(s)G_2(s)$$

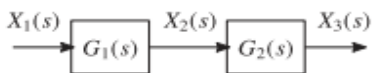
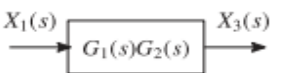
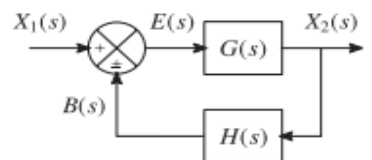
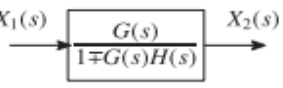
$$\frac{C(s)}{R(s)} = \frac{G_1(s)G_2(s)}{1 + H_1(s)G_1(s)G_2(s) + H_2(s)G_2(s)}$$

# Example: Derive the overall transfer function

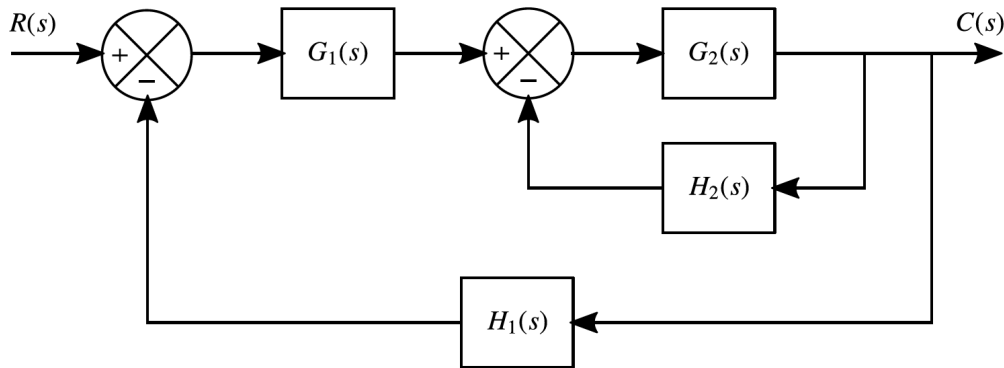


Using Rule No.7:

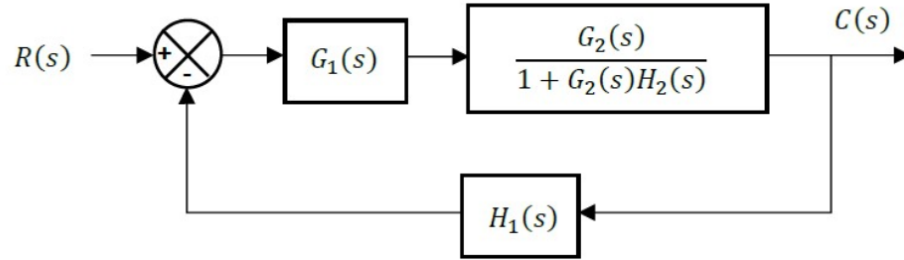


| Rule                           | Original                                                                            | Equivalent                                                                          |
|--------------------------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|
| 1. Cascaded blocks             |  |  |
| 7. Eliminating a feedback loop |  |  |

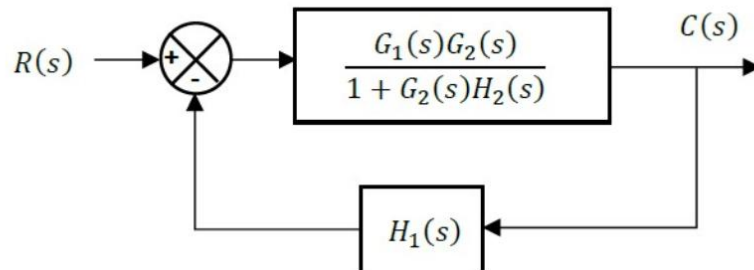
# Example: Derive the overall transfer function

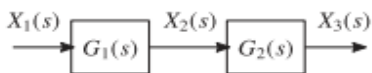
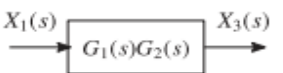
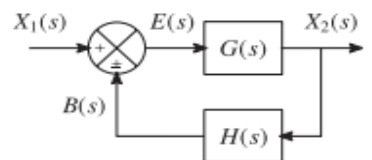
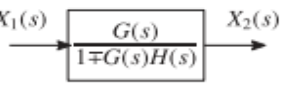


Using Rule No.7:



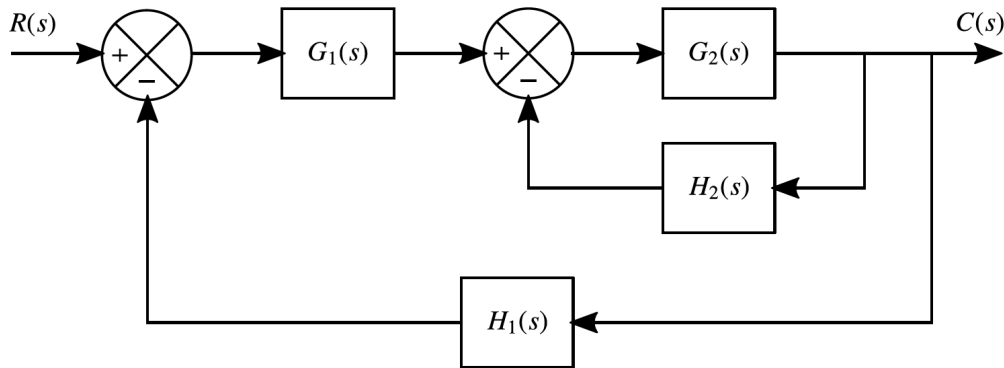
Using Rule No.1:



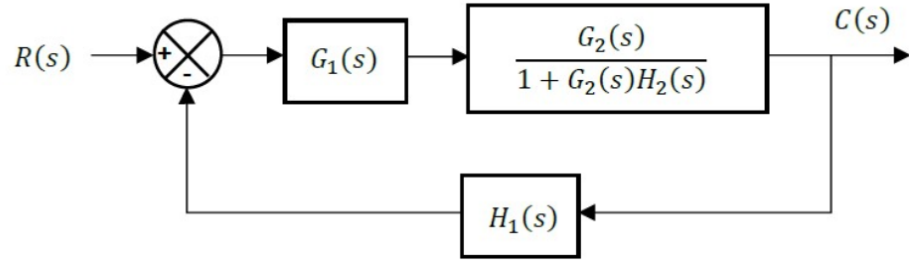
| Rule                           | Original                                                                            | Equivalent                                                                          |
|--------------------------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|
| 1. Cascaded blocks             |  |  |
| 7. Eliminating a feedback loop |  |  |



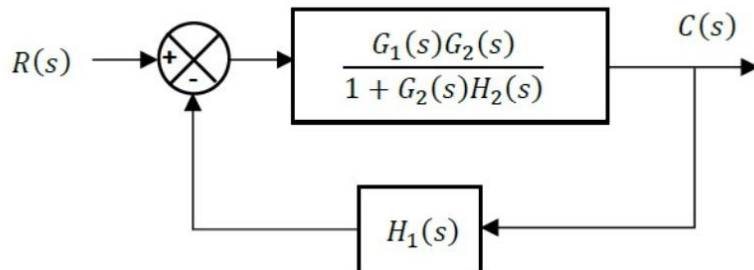
# Example: Derive the overall transfer function



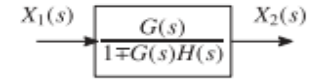
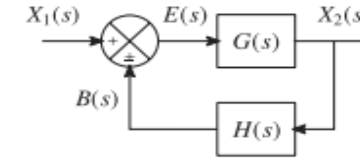
Using Rule No.7:



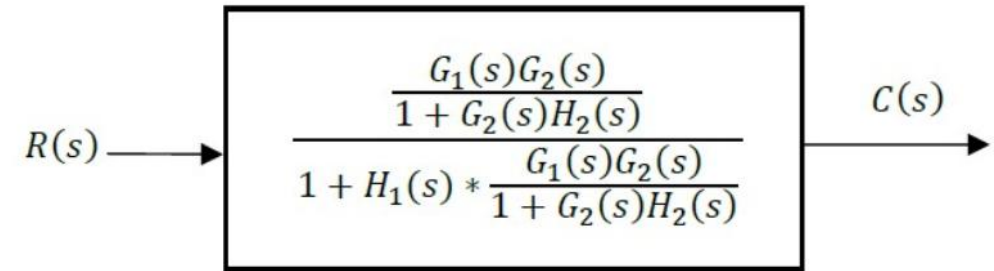
Using Rule No.1:



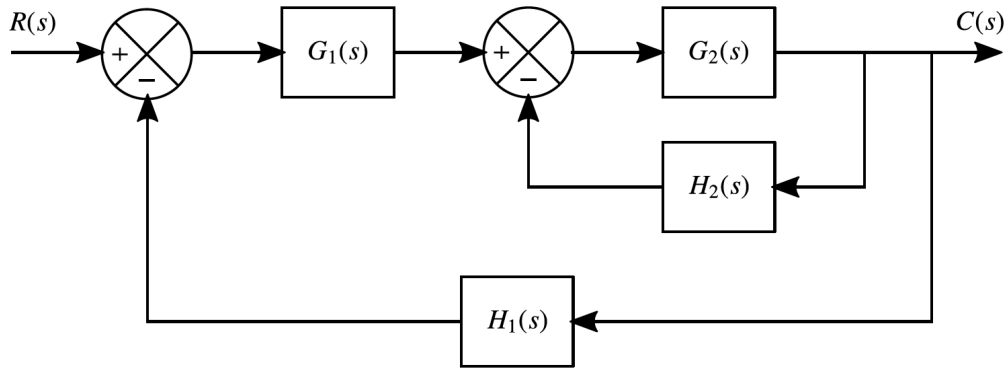
7. Eliminating a feedback loop



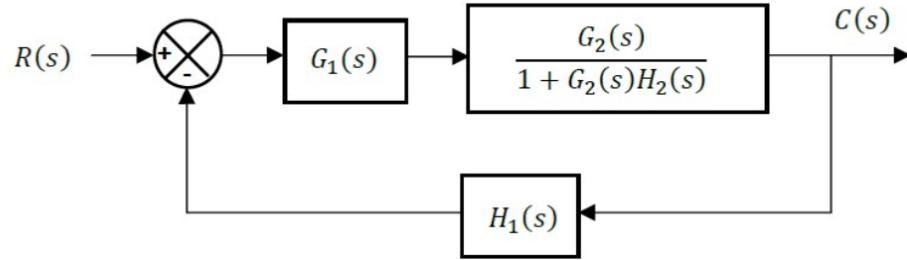
Using Rule No.7:



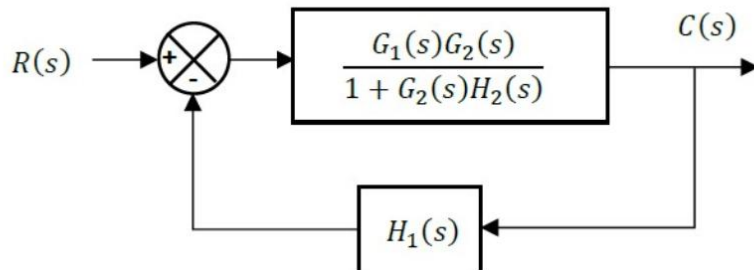
# Example: Derive the overall transfer function



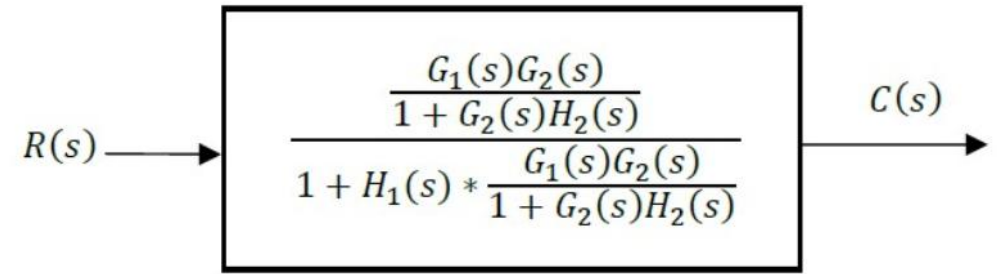
Using Rule No.7:



Using Rule No.1:

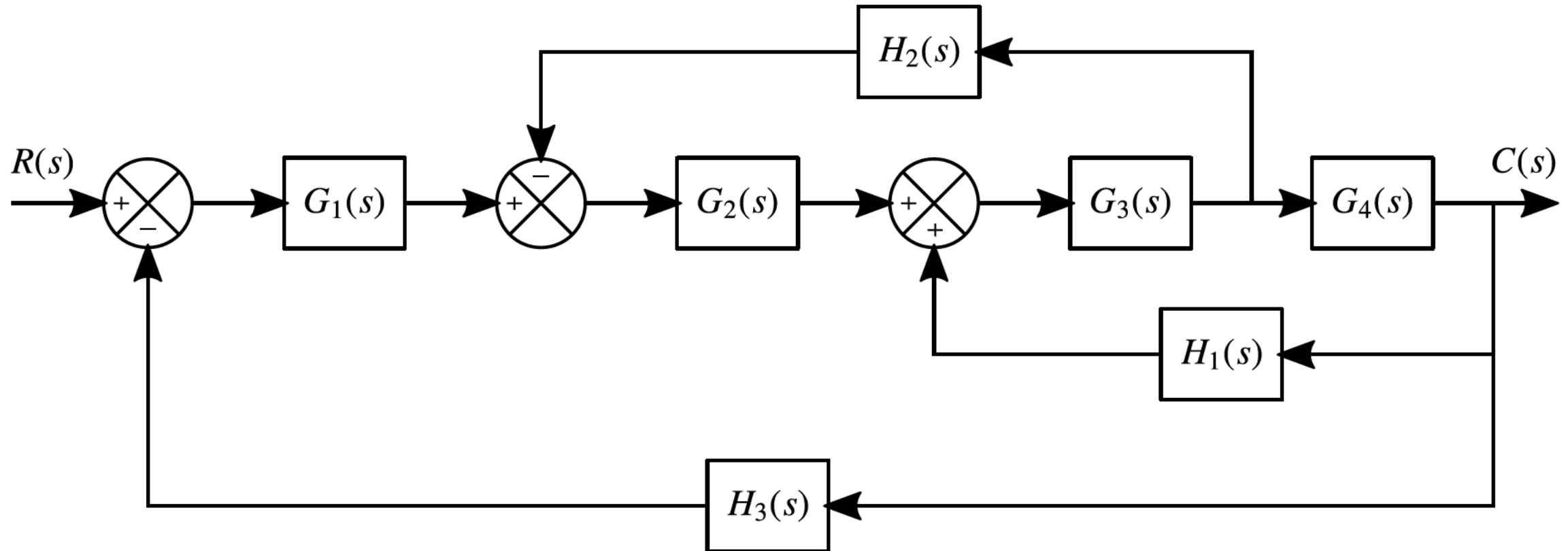


Using Rule No.7:

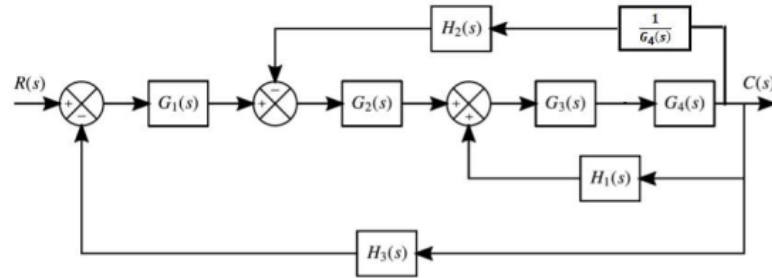


$$\begin{aligned} \frac{C(s)}{R(s)} &= \frac{\frac{G_1(s)G_2(s)}{1 + G_2(s)H_2(s)}}{1 + H_1(s) * \frac{G_1(s)G_2(s)}{1 + G_2(s)H_2(s)}} = \frac{\frac{G_1(s)G_2(s)}{1 + G_2(s)H_2(s)}}{\frac{1 + G_2(s)H_2(s) + H_1(s)G_1(s)G_2(s)}{1 + G_2(s)H_2(s)}} \\ &= \frac{G_1(s)G_2(s)}{1 + G_2(s)H_2(s) + H_1(s)G_1(s)G_2(s)} \end{aligned}$$

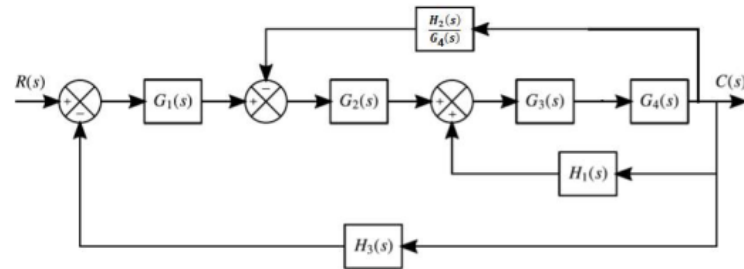
# Exercise: Derive the overall transfer function



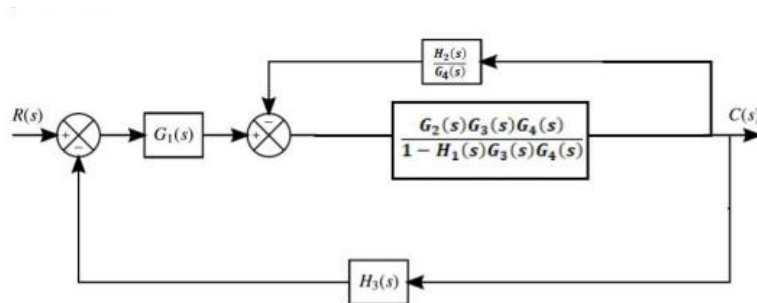
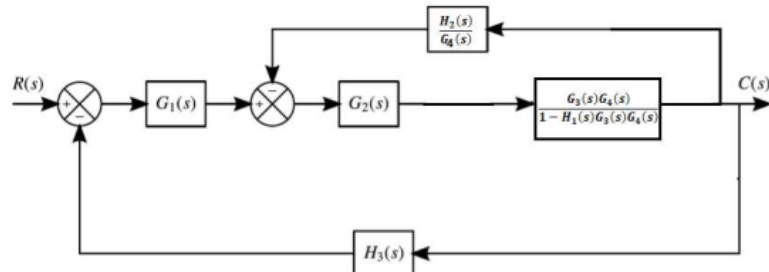
Using Rule No.6:



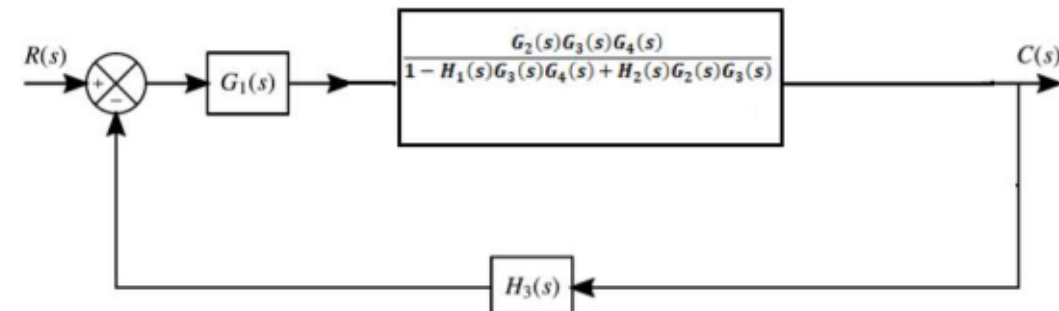
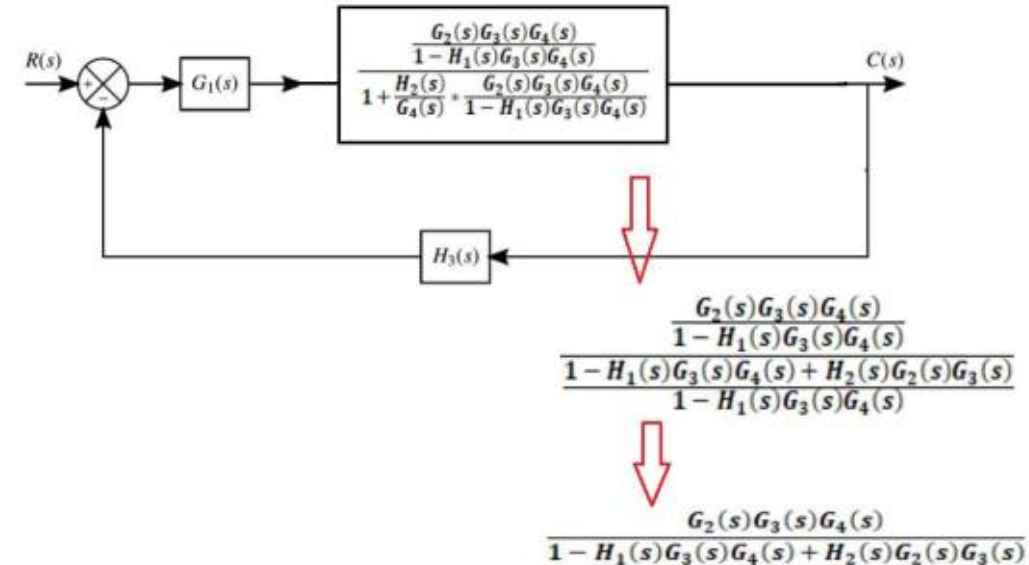
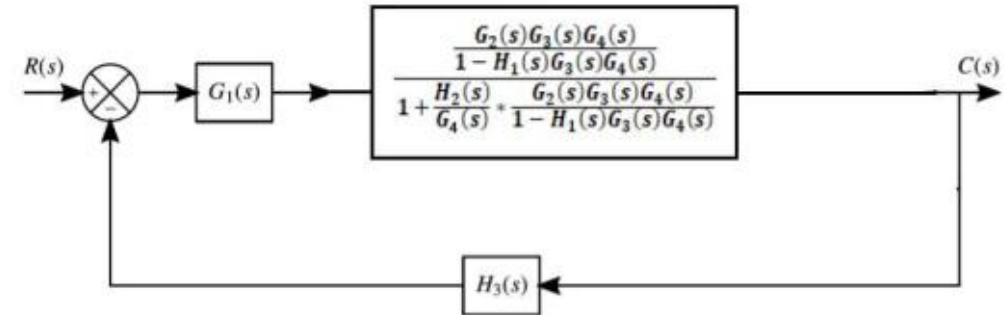
Using Rule No.1:



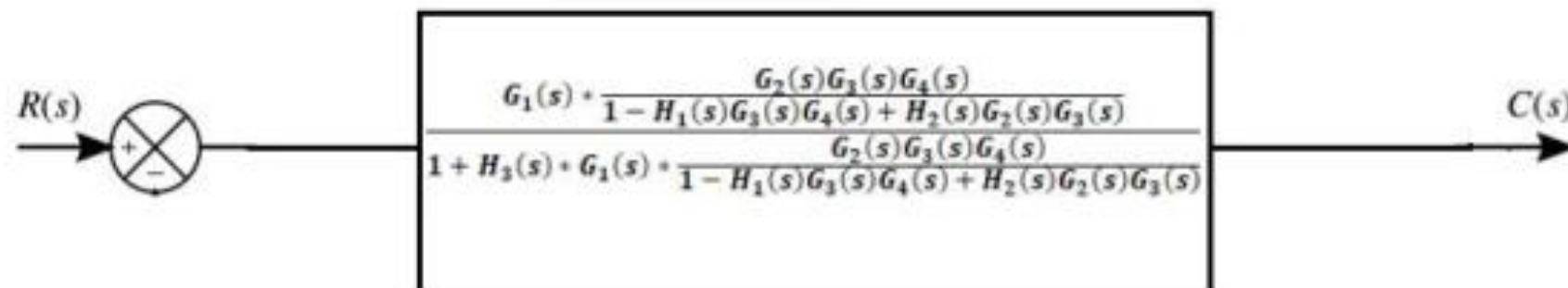
Using Rule No.1&7:



Using Rule No.7:



Using Rule No.1&7:

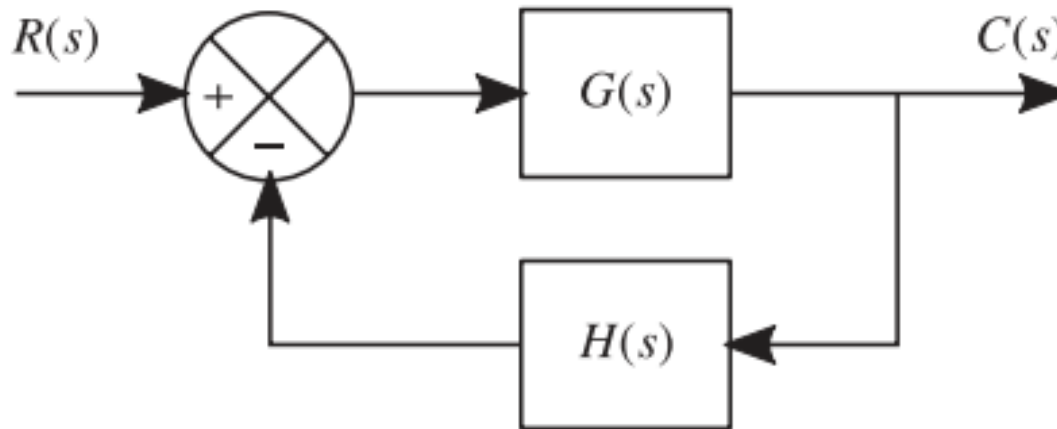


Thus,

$$\begin{aligned}
 \frac{C(s)}{R(s)} &= \frac{G_1(s) * \frac{G_2(s)G_3(s)G_4(s)}{1 - H_1(s)G_3(s)G_4(s) + H_2(s)G_2(s)G_3(s)}}{1 + H_3(s) * G_1(s) * \frac{G_2(s)G_3(s)G_4(s)}{1 - H_1(s)G_3(s)G_4(s) + H_2(s)G_2(s)G_3(s)}} \\
 &= \frac{\frac{G_1(s)G_2(s)G_3(s)G_4(s)}{1 - H_1(s)G_3(s)G_4(s) + H_2(s)G_2(s)G_3(s)}}{\frac{1 - H_1(s)G_3(s)G_4(s) + H_2(s)G_2(s)G_3(s) + H_3(s)G_1(s)G_2(s)G_3(s)G_4(s)}{1 - H_1(s)G_3(s)G_4(s) + H_2(s)G_2(s)G_3(s)}} \\
 &= \frac{G_1(s)G_2(s)G_3(s)G_4(s)}{1 - H_1(s)G_3(s)G_4(s) + H_2(s)G_2(s)G_3(s) + H_3(s)G_1(s)G_2(s)G_3(s)G_4(s)}
 \end{aligned}$$

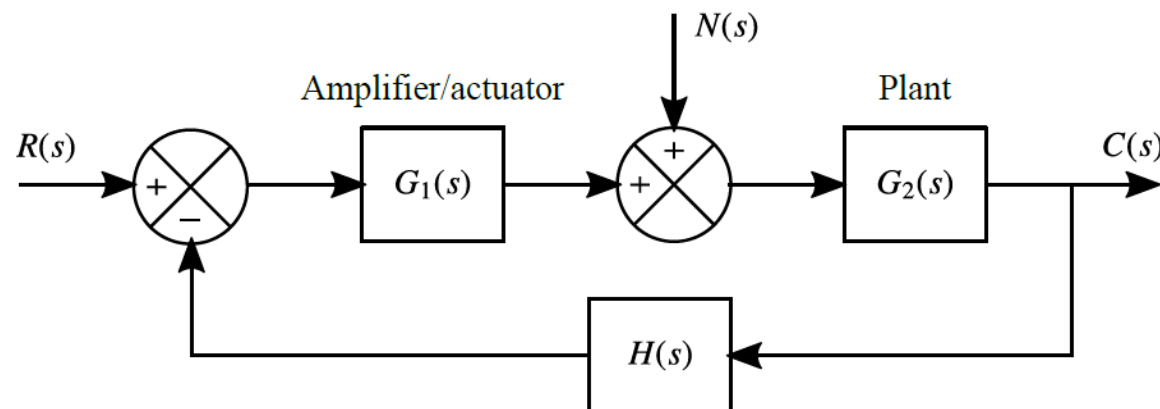
# Closed-Loop Control System Subjected to a Disturbance

- Suppress disturbances in the steady state operation of the plant.
- Disturbance to the plant is incorporated by splitting the transfer function



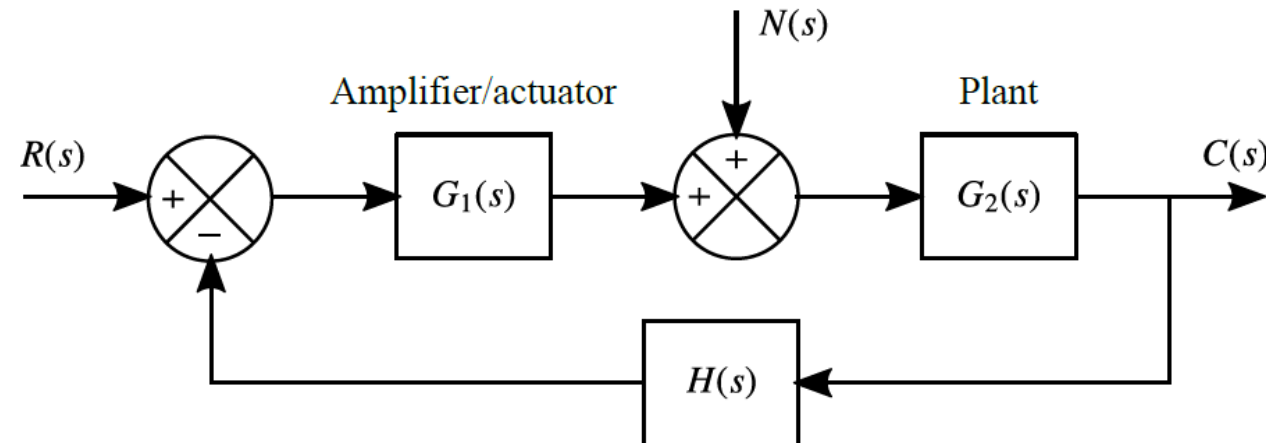
# Closed-Loop Control System Subjected to a Disturbance

- One aim of a control system is to suppress disturbances in the steady state operation of the plant. A disturbance to the plant is incorporated by splitting the transfer function of the plant and amplifier/actuator, as shown below.
- Here  $N(s)$  is the applied external disturbance,  $G_1(s)$  is the amplifier/actuator transfer function and  $G_2(s)$  is the plant transfer function.



# Closed-Loop Control System Subjected to a Disturbance

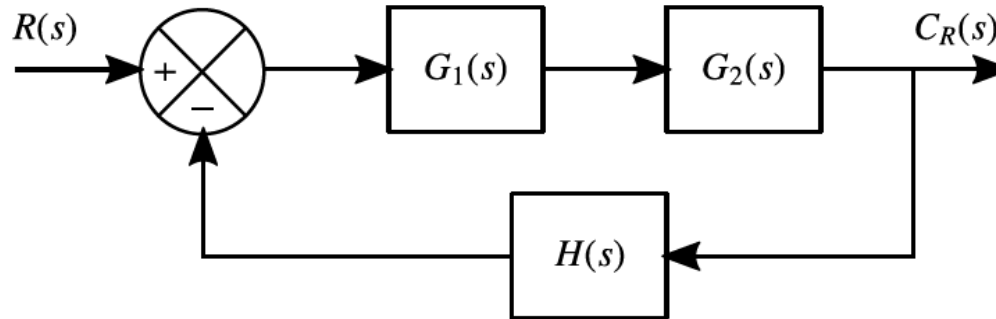
- For **linear systems**, the principle of superposition holds, i.e. if the response to concurrent control inputs and disturbances is required, they can be evaluated individually, then summed together.



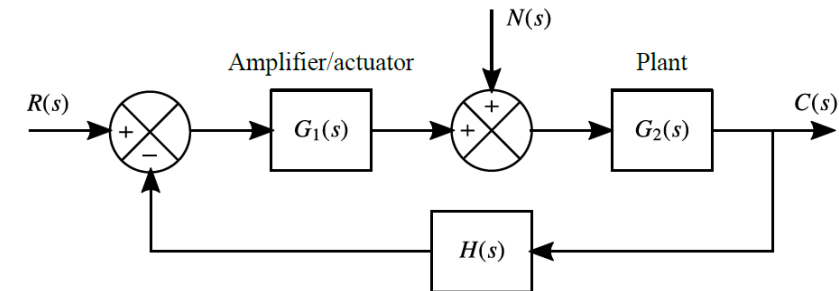


# Closed-Loop Control System Subjected to a Disturbance

- To obtain the transfer function from reference to output, consider the situation where there is no external disturbance (i.e.  $N(s) = 0$ ).

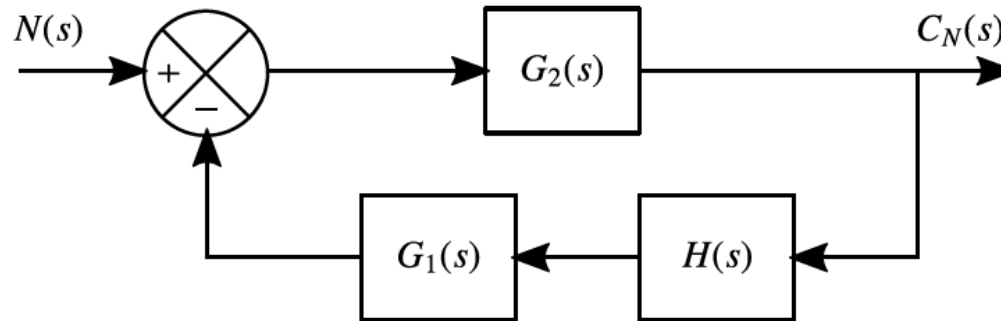


$$\frac{C(s)}{R(s)} = \frac{G_1(s)G_2(s)}{1 + G_1(s)G_2(s)H(s)}$$

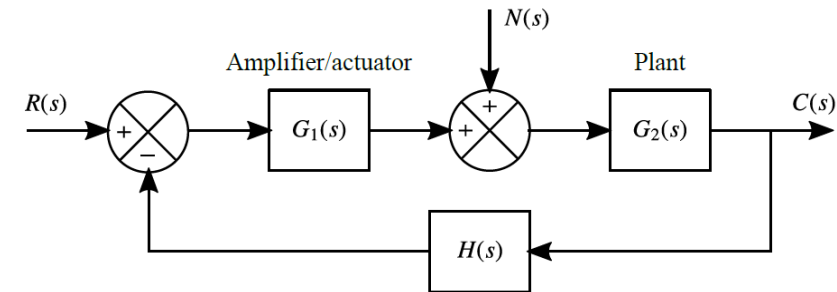


# Closed-Loop Control System Subjected to a Disturbance

- To obtain the transfer function from disturbance to output, consider the situation where there is no reference input (i.e.  $R(s) = 0$ ). The block diagram is then redrawn as below.



$$\frac{C(s)}{N(s)} = \frac{G_2(s)}{1 + G_1(s)G_2(s)H(s)}$$



# Closed-Loop Control System Subjected to a Disturbance

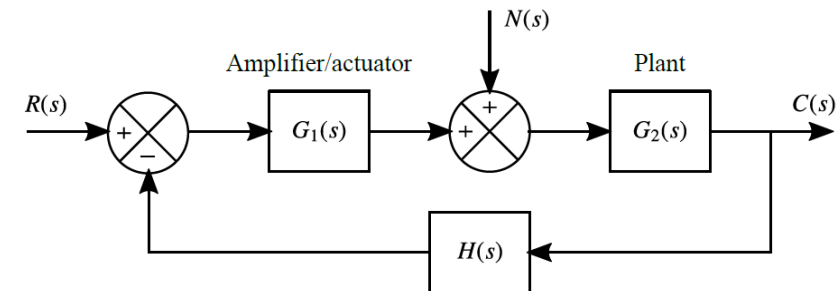
- To obtain the overall response  $c(t)$  we combine both transfer functions:

- $C(s) = C_R(s) + C_N(s)$

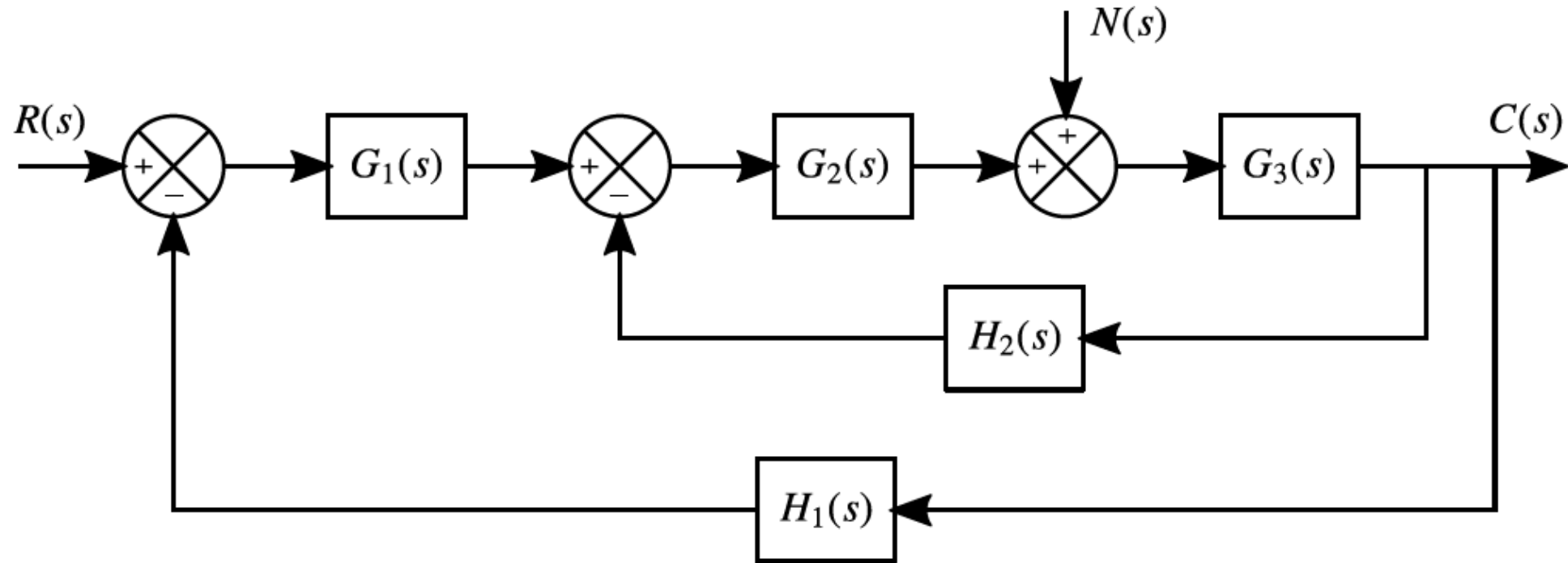
- $$C(s) = \frac{G_1(s)G_2(s)}{1+G_1(s)G_2(s)H(s)}R(s) + \frac{G_2(s)}{1+G_1(s)G_2(s)H(s)}N(s)$$

- $$C(s) = \frac{G_1(s)R(s) + G_2(s)(R(s) + N(s))}{1 + G_1(s)G_2(s)H(s)}$$

- Note that in both cases the denominator has the same value!



# Exercise - Derive the overall transfer function



# Summary

- A complex block diagram can be reduced to describe the system as a single transfer function
- Using superposition, the transfer function for the input and disturbances can be derived

## References:

- Control Systems Engineering, 7th Edition, N.S. Nise
- UESTC3001 2019/20 Notes, J. Le Kernec